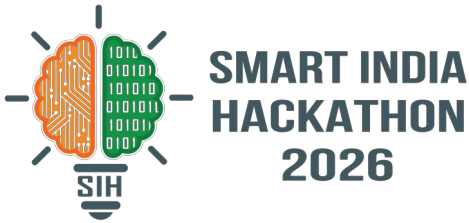


# SMART INDIA HACKATHON 2026



## RASTA AI

Route Intelligence for Essential-Supply Logistics in North Eastern India

SIH26002

PROBLEM STATEMENT ID  
MDoNER · Smart Automation · Software

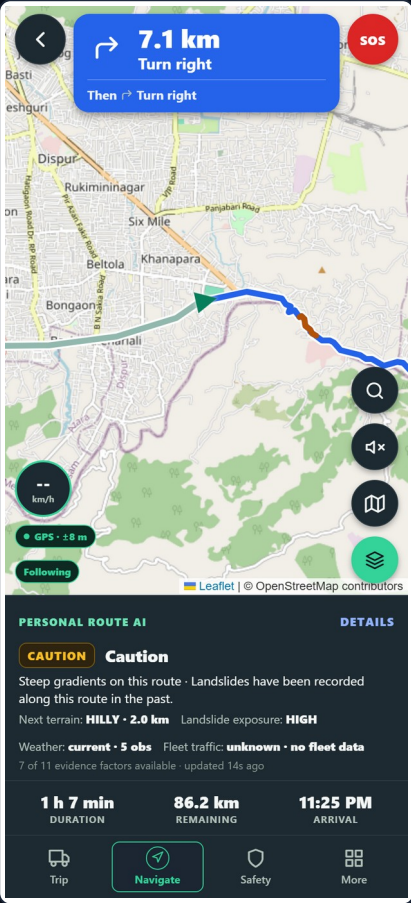
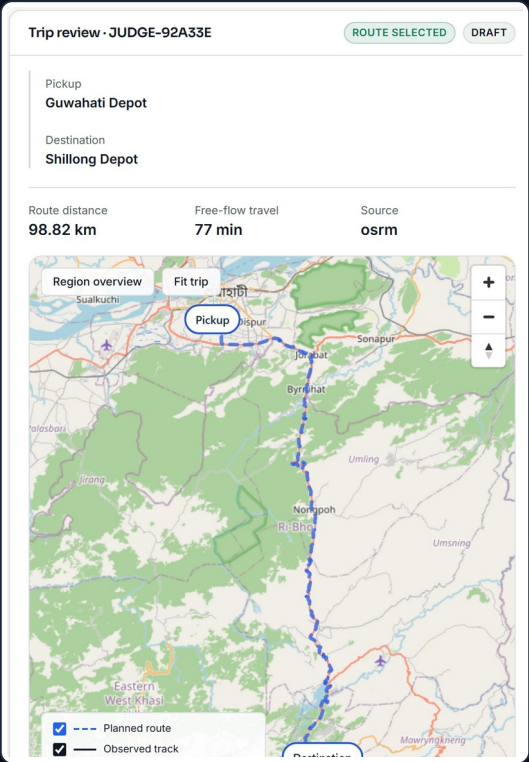
|                         |                                                                                                 |
|-------------------------|-------------------------------------------------------------------------------------------------|
| PROBLEM STATEMENT TITLE | AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER) |
| THEME                   | Smart Automation                                                                                |
| PS CATEGORY             | Software                                                                                        |
| TEAM ID                 | 17 (internal group no.)                                                                         |
| TEAM NAME               | NER-AI LOGISTICS                                                                                |

## SHORTEST IS NOT ALWAYS USABLE.

“Can this essential-supply truck reliably use this corridor now?”

End-to-end working prototype · physical-phone validated · live demo corridor Guwahati → Shillong

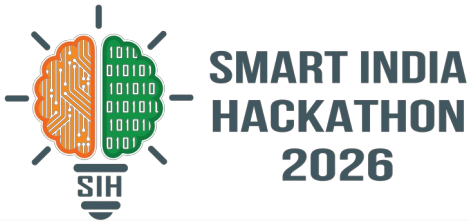
LIVE WORKING SYSTEM manager console + Android driver app



Manager route review · Driver navigation — real screens, real road



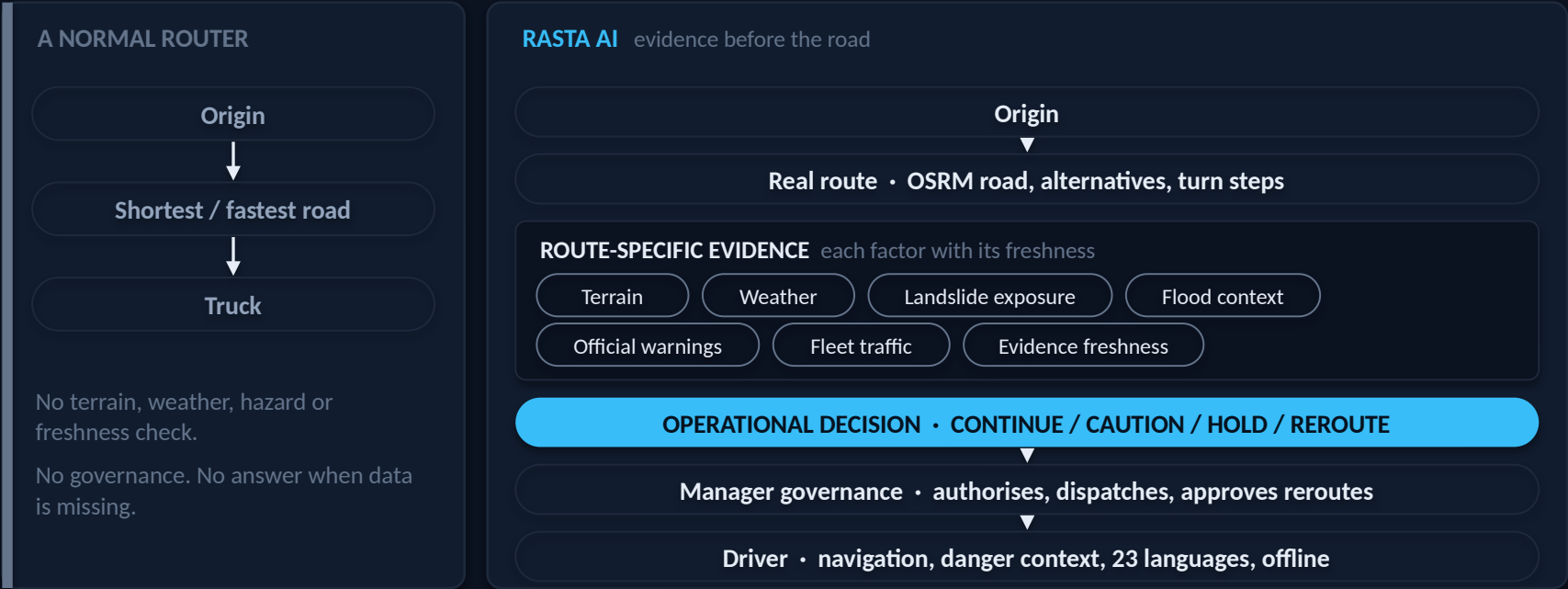
# RASTA AI — ROUTE INTELLIGENCE



PROPOSED SOLUTION

## NOT THE SHORTEST ROAD. THE ROAD THAT IS USABLE NOW.

NER corridors close without notice: monsoon rain, landslides, floods, official warnings, single-lane hill roads, no phone signal. A shortest-path router still sends the truck.



**VERIFIED DATA → DECISION → DRIVER** provider adapters with health + freshness**WEATHER**Open-Meteo · MET  
Norway**TERRAIN**Copernicus DEM ·  
OpenTopoData**WARNINGS**

NDMA SACHET (CAP)

**FLOOD**

GloFAS

**LANDSLIDE HISTORY**

NASA GLC 2007–17

**TRAFFIC**

fleet GPS fixes

**ROUTE-SPECIFIC EVIDENCE · FastAPI backend**

every factor sampled along the selected road · PostgreSQL + PostGIS (Supabase) · OSRM routes, alternatives, turn steps

**DETERMINISTIC SAFETY POLICY**

11 factors · reason codes · UNKNOWN ≠ SAFE · reviewer authorisation when hazard data is missing

**CONTINUE****CAUTION****HOLD****REROUTE****MANAGER REVIEW****DRIVER NAVIGATION****ROUTE-AHEAD MONITOR (60 s)****CONDITIONS CHANGE?****REASSESS / REROUTE**

Re-scored every 60 s along the road ahead; a material change goes back through the policy and a reroute is proposed to the manager — never applied silently.

**BUILT WITH****CLIENTS** Manager console — React + TypeScript, manager accounts only · Driver app — React Native / Expo, Android; one login, server-decided role, offline trip package**STACK** FastAPI · PostgreSQL / PostGIS (Supabase) · React · React Native / Expo · OSRM · OpenStreetMap · Gemini**WHO DECIDES** AI is supporting**DETERMINISTIC SAFETY ENGINE**

makes every operational decision

**GEMINI / OPENROUTER (LLM)**

explains the decision in the driver's language — never decides

**EXPERIMENTAL ML**

research channel only; held outside routing by the safety gate

**GOVERNANCE CHAIN****Verified evidence****Deterministic safety policy****Manager governance****Driver action**

## NOT A CONCEPT. A WORKING END-TO-END PROTOTYPE.

PLAN

REVIEW

DISPATCH

ACCEPT

TRUCK VERIFY

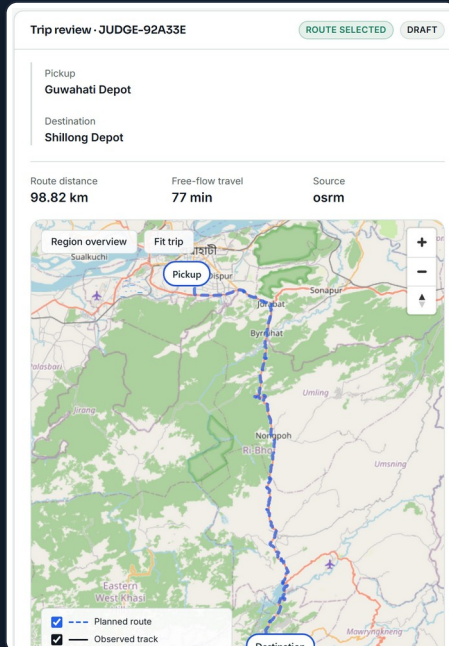
NAVIGATE

OFF-ROUTE

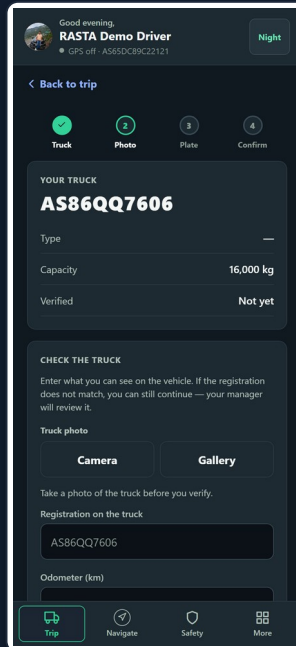
MANAGER REROUTE APPROVAL

DELIVERY

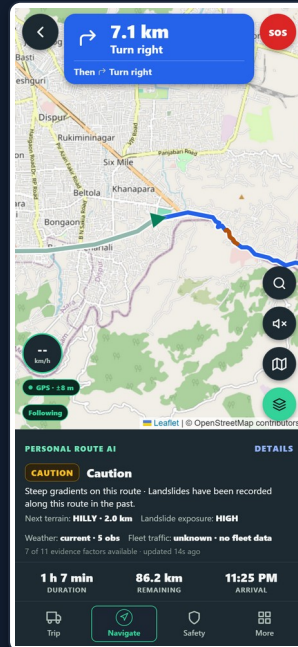
## REAL SCREENS FROM THE CERTIFIED RUN



Manager · route review



Driver · truck check



Driver · navigation

## PROOF physical Android phone · hosted backend · real road

The whole lifecycle above was run end to end on a physical Android phone against the hosted backend, on the real road.

12/12 physical judge-flow steps

9/9 physical role-flow steps

1138+ backend · 621+ driver · 170+ manager tests

PHYSICAL ANDROID · CERTIFIED

Canonical demo Guwahati → Shillong · ≈ 98.8 km real route

KNOWN LIMITS → MITIGATION Render cold start → warm /health before the demo · provider rate limits → cached evidence + explicit UNKNOWN · background push needs Firebase config → in-app alerts already work · experimental ML → outside routing until the gate passes

RASTA AI does not ask only “Which road is shortest?” It asks “Can this truck reliably use this corridor now?”

**FOOD • MEDICINE • FUEL • RELIEF SUPPLIES** — essential logistics for hill communities, decided on evidence

## WHO BENEFITS

### FLEET MANAGERS

Route usability and uncertainty visible before dispatch; auditable decisions.

### DRIVERS

Safer navigation, danger context, emergency support, 23 languages.

### REMOTE COMMUNITIES

More resilient access to food, medicine, fuel and relief supplies.

### GOVERNMENT / OPERATIONS

Auditable route evidence and disruption visibility per corridor.

## DEMONSTRATED TODAY

**23** selectable languages, explicit fallback status

**11** route factors in every decision

**60 s** route-ahead monitor while driving

**12/12** judge-flow steps certified on a physical phone

**100 %** of reroutes reviewed by a human

**0** unknown factors counted as safe

## HUMAN GOVERNANCE AI has no uncontrolled authority

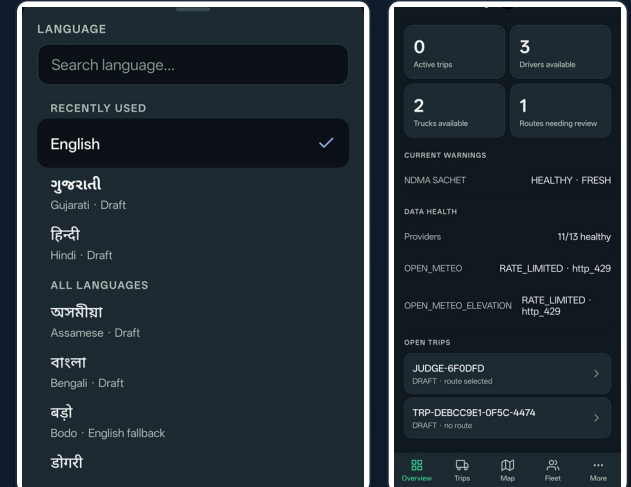
Verified evidence

Deterministic safety policy

Manager governance

Driver action

## REAL SCREENS • APK 1.0.18



Driver • 23 languages

Manager • mobile

Real screens, APK 1.0.18. One login, server-decided role: a manager gets a mobile fleet view, a driver gets navigation. Each language shows its status — Verified, Draft or English fallback — so nobody is misled.

## RESEARCH / EVIDENCE SOURCES BEHIND THE ROUTE DECISION

## ROUTING / PLACES

**OSRM** • OpenStreetMap • Nominatim

real road geometry, alternatives and turn steps; place search on the corridor

## OFFICIAL / FLOOD

**NDMA SACHET (CAP)** • Copernicus GloFAS

official warnings polled with freshness; river discharge against its 30-day mean

## AI ASSISTANCE

**Google Gemini**

explains the decision in the driver's language — never safety authority

## WEATHER / TERRAIN

**Open-Meteo** • MET Norway • Copernicus DEM • **OpenTopoData**

forecast sampled along the road; climb, gradient and steep segments (GLO-90)

## LANDSLIDE HISTORY

**NASA Global Landslide Catalog**

historical exposure; inventory 2007–2017, ≤5 km location-accuracy filter

## PROBLEM STATEMENT

**MDoNER** • SIH26002

AI-based smart logistics and accessibility intelligence for the NER

## EXPERIMENTAL LANDSLIDE RESEARCH

95 %

## RECALL

on the NER geographic holdout — 154 catalogued events, no NER row used in training

## SAFETY GATE

**NOT DEPLOYED****FPR 50 %** too high for production deployment

TRAIN

HELD-OUT NER TEST

USEFUL SIGNAL FOUND

FALSE POSITIVES TOO HIGH

SAFETY GATE

NOT DEPLOYED

**VALIDATION BEFORE AUTOMATION.** Strong recall was not enough. Held-out testing exposed the false-positive rate, so the gate keeps the model out of routing.

Precision 0.32 • F1 0.48 • PR-AUC 0.55 • ROC-AUC 0.83 • Brier 0.12 • calibrated logistic regression on NASA GLC + ERA5-Land rain + DEM slope • docs/HAZARD\_ERROR\_ANALYSIS.md

## REFERENCES

gpm.nasa.gov/landslides (NASA GLC) • sachet.ndma.gov.in • global-flood.emergency.copernicus.eu (GloFAS) • open-meteo.com • api.met.no • opentopodata.org • Copernicus DEM • project-osrm.org • openstreetmap.org • github.com/nxtlucifer/ner-ai-logistics (repository, tests, model registry)